

Global Industry Challenge 2026 *Phase 3 Challenge Description*

CHALLENGE PROVIDER:

Mitsubishi Chemical Group and the National Institute of Advanced Industrial Science and Technology (AIST)

FOCUS AREA:

Advanced Materials

TITLE:

Quantum Materials Discovery Challenge: Scaling Generative Quantum Eigensolver (GQE) Using NVIDIA CUDA-Q

1. INTRODUCTION

Welcome Message

Mitsubishi Chemical Group and the National Institute of Advanced Industrial Science and Technology (AIST) are advancing next-generation materials discovery through the integration of computational science, artificial intelligence, and emerging quantum technologies.

Accurate simulation of quantum systems is a central challenge in chemistry and materials science. As system size grows, classical computational methods face exponential scaling limitations, restricting their applicability to industrially relevant problems.

This challenge aims to explore how hybrid quantum-classical approaches—particularly generative quantum algorithms—can overcome these limitations and unlock scalable quantum simulation capabilities.

Overview of the Organizations

Mitsubishi Chemical Group is a global leader in advanced materials and chemical innovation, headquartered in Tokyo, Japan. The company operates across a wide range of sectors—including performance materials, specialty chemicals, healthcare, and sustainable solutions—serving industries such as electronics, mobility, energy, and life sciences. Mitsubishi Chemical Group is actively investing in next-generation research capabilities, including the integration of artificial intelligence and quantum computing to accelerate materials discovery and optimize complex chemical processes. In this Challenge, Mitsubishi Chemical Group brings deep domain expertise in industrial materials science and real-world application needs, helping to guide the development of scalable quantum simulation approaches with tangible commercial impact.

The National Institute of Advanced Industrial Science and Technology (AIST) is one of Japan's largest public research organizations, dedicated to bridging the gap between scientific discovery and industrial application. With a broad research portfolio spanning energy, materials, electronics, artificial intelligence, and quantum technologies, AIST plays a central role in advancing innovation that supports economic and societal development. The organization is a global leader in computational science and materials modeling, with strong capabilities in high-performance computing and emerging quantum methodologies. In this Challenge, AIST contributes cutting-edge research expertise and technical leadership in quantum simulation and hybrid quantum-classical algorithms, supporting participants in developing scalable solutions for complex materials systems.

2. THE CHALLENGE

Problem Statement

Ground-state energy estimation is a foundational problem in quantum chemistry and materials science. While classical methods struggle with scaling, quantum approaches such as the Variational Quantum Eigensolver (VQE) face challenges related to optimization complexity, circuit design, and scalability.

The Generative Quantum Eigensolver (GQE) introduces a novel paradigm in which a classical generative model produces quantum circuits that approximate ground states. Early demonstrations have shown promising results for small-scale systems (~10–12 qubits), but scaling to larger systems remains a critical open challenge.

Objective

Develop scalable implementations of GQE capable of addressing significantly larger quantum systems (target: ~40 qubits).

Desired Outcomes

- Scalable GQE architectures
- Demonstration at ~40 qubits (ideal) or 20–30 qubits (acceptable)
- Chemical accuracy (~1.6 mHa) where possible
- Efficient hybrid workflows
- Benchmarking vs classical and VQE

3. RESOURCES AVAILABLE

Recommended resources include:

- NVIDIA CUDA-Q
- GPU simulation
- qBraid platform
- Qiskit, Cirq, PennyLane
- PyTorch, TensorFlow
- Public datasets



4. GUIDELINES AND EXPECTATIONS

Submission Requirements

The **official Global Industry Challenge cover page template is required** and must be included as the first page of the submission. Teams may not recreate, modify, or substitute this template. Submissions that do not comply with these requirements may be disqualified.

Your submission should address:

- Focus area and rationale
- Technical approach to quantum integration
- Stakeholder relevance
- Data modeling strategy
- Quantum platform justification and resource needs

Phase 3: Prototype Development

Participants must define:

- Target system
- GQE-based approach
- Data modeling strategy
- Platform and resource estimation
- Hybrid architecture

Evaluation Criteria

Submissions will be scored using a rubric by reviewers. Judges will pay close attention to the following:

1. Scalability (primary)
2. Accuracy (chemical accuracy preferred)
3. Algorithmic innovation
4. Computational efficiency
5. Hybrid system design
6. Benchmarking
7. Clarity and reproducibility

Bonus Points:

- >40 qubit scaling
- Noise-aware design

Category	Description
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1. Scalability (Primary Criterion)	Strength and credibility of the proposed approach to scale GQE or hybrid quantum methods beyond current limits. Includes ability to target larger systems (e.g., 20–40+ qubits), identification of bottlenecks, and strategies to manage increasing system complexity.
2. Accuracy and Scientific Validity	Clarity of expected results and alignment with meaningful scientific benchmarks, including chemical accuracy (~1.6 mHa) where applicable. Demonstrates understanding of trade-offs between accuracy, scalability, and computational cost.
3. Algorithmic Innovation	Originality and technical strength of the proposed GQE or hybrid approach, including generative model design, circuit generation, and optimization techniques. Shows advancement beyond standard methods such as VQE or existing baselines.
4. Computational Efficiency	Effectiveness of the approach in managing computational resources, including circuit efficiency, training complexity, and classical compute requirements. Demonstrates awareness of performance constraints and optimization opportunities.
5. Hybrid System Design	Quality of integration between quantum and classical components, including workflow design, division of tasks, and overall system architecture. Reflects a practical and well-structured hybrid approach.
6. Platform Use	Expanded appropriateness of selected QC or simulator, qubit/depth/shot estimates, and integration.
7. Phase 3 Execution	Detailed write-up of concrete results, observations and conclusion. <i>This is a critical Phase 3 judging criteria.</i>
8. Clarity of Communication	Structure, coherence, and completeness of the 3-page technical submission for a mixed audience of quantum researchers, materials scientists, and industry stakeholders

Rules and Regulations

- Solutions must be reproducible. Code, data references, and run instructions must be sufficient for a third-party reviewer to verify the headline results in Phase 3.
- Teams must disclose any use of proprietary data and cannot submit results that depend on data the judges cannot access.
- Use of generative AI tools is permitted for code support and writing, but the technical contributions, formulations, and results must be the team's own work.
- Any siting or planning recommendations are research outputs and do not constitute investment, regulatory, or operational advice.
- Teams should be careful with grid topology and asset-level data that may be sensitive, and should default to publicly released test systems wherever possible.

PHASE 3 SUBMISSION REQUIREMENTS

Phase 3 requires working implementations run on QCI's Dirac-3 via qBraid with measurable outputs on the three key performance metrics. All three pipeline stages must be executed on a concrete grid instance, and results must be reproducible by judges.

Phase 3 moves from conceptual design to executed, reproducible results. The following requirements apply to all submissions.

Submission Format

Upload a single zipped folder named TeamName_Challenge_Phase3.zip to Aqora. The zip file must contain:

- **Write-Up:** Maximum 5 pages PDF (not including references or cover page), using 11-point Times New Roman font and single spacing. Review the Aqora Challenge pages for the specific instructions on the criteria.
- **Source Code Folder (if applicable):** Include all code files required to reproduce your results. Organize by module if applicable. Ensure scripts are well-commented and executable on qBraid without external configuration.
- **README.md File (if applicable):** The README.md should clearly describe: Team Name, Project title, and challenge track; setup instructions (e.g., environment dependencies, packages); step-by-step instructions to run the code on qBraid; expected inputs/outputs; known limitations or assumptions. Include a 'Launch on qBraid' button: <https://github.com/qBraid/community/discussions/3>

Judging Criteria Alignment

- Verify results are successfully run on qBraid, and judges can re-run without modification.
- Ensure results run by judges match the performance and results described in the write-up.
- Include clear documentation and instructions in the README.md for reproducibility.

Failure to meet these criteria could result in disqualification.

SUMMARY: TOP ACTIONS FOR PHASE 3

Regardless of which challenge track you are competing in, the following actions will most directly improve your Phase 3 score:

- **Run your code (if applicable) on qBraid, QCi, D-Wave, or IBM before submission:** Judges will re-run your code exactly as submitted. A result that cannot be reproduced will not receive credit for that result.
- **Report a classical baseline:** Every challenge requires a comparison against a non-quantum method on the same problem instance. This is the single most common gap in Phase 2 submissions.
- **Demonstrate and explain quantum advantage:** Develop your Phase 2 platform and resourcing plan to demonstrate how you used QC or simulators to provide benefits over classical approaches. Expanded appropriateness of selected QC or simulator, qubit/depth/shot estimates, and integration. Detailed write-up of concrete results, observations, and conclusions. *This is a critical Phase 3 judging criterion.*
- **Be specific about numbers:** Present your results. Qubit count, circuit depth, shot budget, wall-clock runtime, and key metric values must appear in your write-up. Qualitative descriptions of performance are not sufficient.
- **Write a complete README.md:** The README is the first thing judges read. A clear, step-by-step README that lets a judge reproduce your result is worth more than an extra paragraph of theoretical justification in the write-up.
- **Be honest about limitations:** Every challenge rubric rewards understanding of where the quantum approach does and does not provide benefits. A clear discussion of limitations paired with a working implementation will outscore an overstated claim paired with a partial one.